5.1. INTRODUCTION

Transistors are the fundamental building block of modern electronic devices basically used for controlling, amplifying and generating electrical signals. When a third doped element is added to a crystal diode in such a way that two PN junctions are formed, the resulting device is known as transistor. The transistor is an entirely new type of electronic device, capable of achieving amplification of weak signals.

Who invented the transistor?

The transistor was invented by William Schockley, John Bardeen, Walter Brattain.

History of Transistor

The transistor was first developed by John Bardeen, Walter Brattain and William Shockley at Bell Labs in December 1947. This new device has characteristics that could overcome many of the fundamental limitations of vacuum tubes. Transistors have very long life, small in size, light weight and mechanically rugged and required no filament.



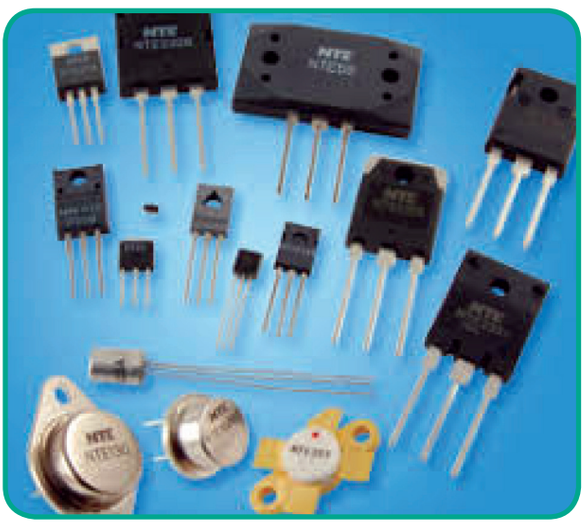
William Shockley

Initially, transistors were made from semiconductor material called Germanium. Further, silicon has been used as a semiconductor material, since silicon has superior performance at higher operating temperature. It is now widely used.

5.2. TRANSISTOR

It is made up of semiconductor material such as Si and Ge. Usually, it comprises of three terminals namely, base, emitter and collector for providing connection to the external circuit. Today, some transistors are packaged individually and many transistors are fabricated according to the design of an embedded integrated circuits.

A transistor is a semiconductor device used to amplify or switch electronic signals and electrical power.



Types of Transistors

Types of transistors

1. Bipolar Junction Transistor (BJT)
2. Field Effect Transistor (FET)
3. SMD Transistor
4. DIAC, TRIAC and IGBT.

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